

DTC 2 Project Report

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106-2: Design Thinking & Communication

Executive Summary

WedgeRise: A Stabilizer and Height Adjuster for Dorm Rocking Chairs

Problem Statement

Many dormitory chairs used across U.S. college campuses, including at Northwestern University, are prone to instability due to their tubular rocking-leg design. Students frequently report tipping over backwards, falling from standing position, scraping floors, and feeling discomfort due to insufficient seat height.

[visual on the problem]

Existing commercial solutions fail to address the unique geometry of these chairs, prompting the need for a custom product. The purpose of this project is to design a safe, stable, and height-adjustable attachment for dorm chairs with tubular legs, retrofitted to existing chairs without power tools and maintaining a minimal footprint.

Methodology

Our design process was grounded in user-centered research, benchmark testing, ideation, prototyping, testing, and iteration. First, we gathered over 110 unmet needs and systematically prioritized them based on important criteria. After selecting the dorm chair issue, we conducted patent and market research to improve on ideation, and developed multiple prototypes and mockups. Finally, we tested them with users including stability under force using push and pull force gauge, and feedback on comfort and safety, to guide our redesigns.

Design Summary

The resulting solution, WedgeRise, is a spring-loaded wedge system that attaches to the rear legs of dorm chairs via steel hose clamps.

[visual of the whole design system]

It stabilizes the chair during backward lean by dampening impact forces and increasing resistance to tipping, while moving the tipping point further back. Interchangeable wooden riser blocks allow users to select between baseline and elevated seating configurations, ensuring ergonomic flexibility for different body types and desk heights.

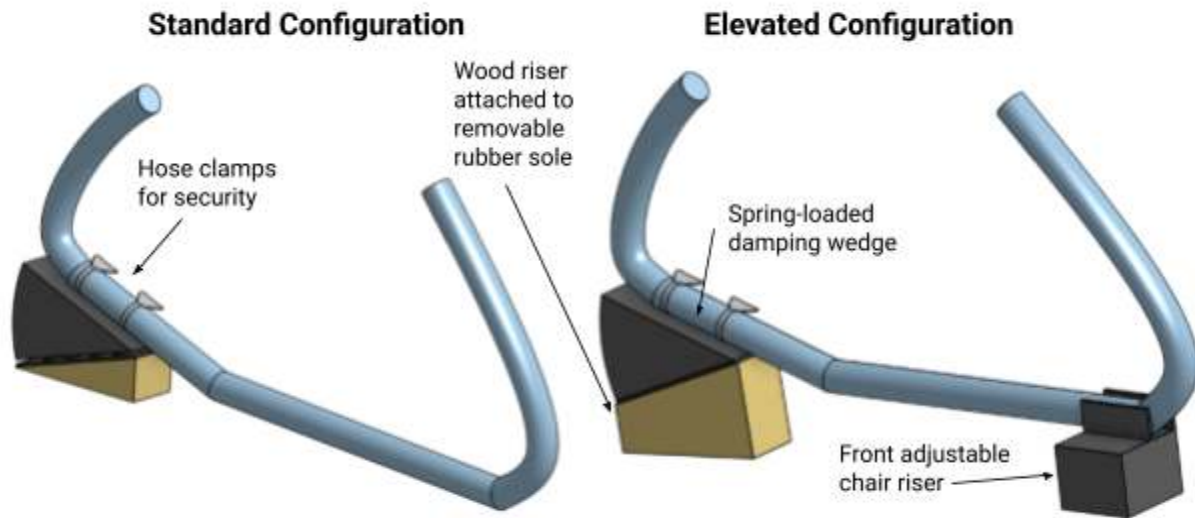


Figure 2: Design

Future Developments

WedgeRise significantly improves chair stability and user safety, while remaining compact, affordable, and tool-free to install. Future development should explore fully adjustable telescoping mechanisms, broader compatibility for non-cylindrical legs, and testing with more diverse user groups beyond Northwestern's student body.

Introduction

Problem Statement

Most college dorms have rocking chairs that the school invested money into. Many users of these chairs that our team surveyed have complained of falling backwards out of their seat while doing homework, or nearly slipping over the chair while trying to reach a poster on their wall. Others have even complained of discomfort due to the low height. This is not just an issue for Northwestern dorms, but also most college students in the US.

Although we had many great options from our need-finding research, we ultimately decided to focus on the issue of Northwestern's dorm chairs. When we looked at results from our surveys and interviews, we found that people had a plethora of complaints about these chairs. Their lack of stability and high tendency to tip back made for a scary and distracting study environment, and some argued that the chair was too low to comfortably write or type on a computer. It's fairly easy to see why these problems occur—the base of these chairs are not completely flush to the ground at all points, making it very easy to lose balance.

While there are many furniture risers and wedges all across the internet, most of these solutions simply won't work because the dorm chairs have two continuous legs instead of four. [expand on existing solutions]

Mission Statement

Design a retrofittable tubular attachment for college students with rocking dorm chairs, who want a stable and height adjustable dorm chair. This product should address issues of rocking while leaning back, base scraping the floor, and safety standing on the chair.

The Wedge Rise

Our design is a spring-loaded wedge which clamps to the back of the chair legs to decrease tipping and help stabilize the chair. There is also an option to raise all four corners of the chair to make the seat higher if users feel more comfortable. Through testing and user feedback, we have made plenty of positive changes to our design to make it as helpful as possible.



Figure _:

This report will take readers through requirements, design rationale, and limitations of our product. We will discuss the users of our design, the criteria our design must meet to perform well, a deeper look into our final prototype and how it works, and future developments that could better our design in the future.

Users & Requirements

Through a problem finding and filtering process, we narrowed the focus to one problem: Improving dorm rocking chairs. This particular problem stood out for several reasons. It appeared three times in our survey responses, and ranked #2 overall in our problem scoring matrix. It demonstrated high feasibility within the constraints of our course timeline. It serves a large and relevant user base as shown by the ample online forum response—Northwestern students in dormitories, as well as most college dorms in the US express frustration about these rocking chairs being dangerous. Finally, our patent research and commercialized searches showed a lack of adequate existing solutions.

Primary Users:

- Northwestern students who live in dorms with the rocky, tubular base chairs.
- Other college students who sit in chairs with a similar stature and swaying capabilities

Secondary Stakeholders and Users:

- Friends, family, or acquaintances of the students who will sit in the chairs.

Illustrative User Scenario

This scenario is based on the stakeholder interview, in-person observation, and personal experience from team members.

After a long day filled with classes and work, Bobby, a first-year student at Northwestern University, finally returns to his dorm room hoping to relax and complete tedious class assignments. As he settles into his chair, he notices scratch marks on the floor caused by repeated movement of the chair. Glancing up, he realizes that a poster has fallen from the wall. Attempting to re-stick it, he then stands on the unstable chair in an effort to reach higher, but loses his balance, falls, and injures himself against the desk.

Frustrated and in pain, Bobby sits back in his chair to recover. However, the chair, lacking proper support and stability, tips backward, causing him to fall again—this time striking the wall behind him. Now physically hurt, emotionally drained, and overwhelmed by the cumulative stress of the day, Bobby is going nuts.

Requirements

To create a design that truly meets the needs of our users, we carefully considered every aspect of the dorm chair experience, with a focus on improving each area. By reflecting on the challenges our users face, we aimed to design a retrofittable tubular attachment for college students with rocking dorm chairs, who want a stable and height adjustable dorm chair. This

product should address issues of rocking while leaning back, base scraping the floor, and safety standing on the chair.

Table 1: Requirements for the scope of the project

Needs	Metrics	Units	Ideal Value	Allowable Value	Justification
Durable and low maintenance	Lifespan before structural failure or performance degradation	Years	15	5	College furniture is typically used over multiple student cohorts. Industry furniture (e.g., Herman Miller) sets lifespan targets of 12 years [1]. Given expected rough use in dorms, 5 years is acceptable for prototype
Retrofittable	Ease of installation by user without specialized tools	1-5	1	3	Based on user testing feedback, most students do not have access to power tools. A tool-free installation (rating 1) is preferred. Installation steps > 3 were found to reduce user compliance.
Size	Additional floor footprint introduced	ft ²	0	1	Dorm rooms are typically under 150 ft ² . Maximizing compactness minimizes obstruction. Ideal footprint aims for no added protrusion beyond existing chair.
Height adjustability	Range of vertical adjustment possible for chair seat	in	>5	>2	ADA recommendations for accessible seating suggest height flexibility to accommodate users of varying sizes [2]. Survey results indicated that students prefer ergonomic adjustment to desk height.
Stability when use	Tipping resistance and	1-5	5	4	Interviews and injury reports indicated a significant user pain

	sway under normal and moderate load				point around tipping (Appendix A). ASTM F2613-21 standard for seating requires minimal deflection during use [3].
Maximum weight capacity	Structural load bearing without deformation or failure	lbs	300+	275+	ANSI/BIFMA X5.1 standard for general-purpose chairs sets the minimum weight limit at 275 lbs. To support a diverse user population, a minimum is set [4].

Design Concept and Rationale

The WedgeRise is a retrofitted attachment for the backs of Northwestern University dorm chairs that prevents the chairs' characteristic rocking nature and provides the stability of a standard four-legged chair for users. Using a spring-loaded doorstop at the top of the product, the WedgeRise also helps significantly reduce the chair's tipping point by dampening the impact of downwards/backwards force upon the chair, drastically reducing users' chances of falling out. The WedgeRise also comes with two different sized attachments: one that makes the product's bottom flush with the front part of the chair legs (making it fully stable), and one that adds two inches of height and is complemented with two bed risers that are attached to the front legs (making the chair taller for shorter users). Both sized wooden bases and the bed risers have felt bottoms to prevent scraping with the floor and enable smooth scooting of the chair.

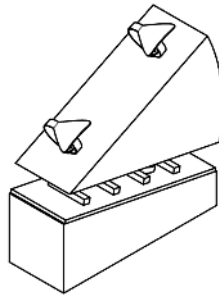


Figure 1: Visualization of our proposed design

Our design (Figure 1) consists of two spring-loaded wedge risers connected to the backs of the chairs via 1.5 inch metal hose clamps. These attachments can then connect to wooden blocks of two sizes, each having a rubber layer on top that fits and locks into indents found on the bottoms of the spring-loaded wedges.



Figure 2: Worm Gear Pipe Clamp

WedgeRise implements metal hose clamps with wing nuts (Figure 2) to help adjust clamp tightness. These were chosen following previous mockups that instead used one-size plastic clamps, which were found to lack durability and to lack enough tightness around the chair leg. Conversely, these metal hose clamps provide both durability and tightness by being made of steel and having a wing nut which the user can tighten onto the chair leg. They were therefore the most optimal clamps to use in ensuring a tight fit alongside longevity.

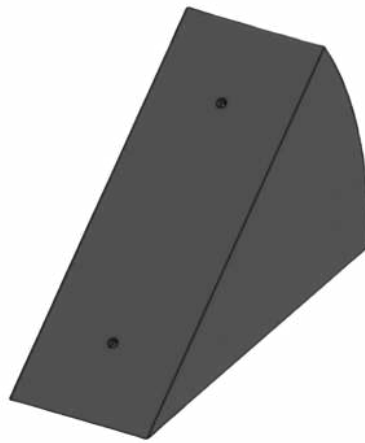


Figure 3: Adjustable door stops used in WedgeRise design

Two metal hose clamps are attached to each adjustable door stop (Figure 3), the initial piece attached to the chair leg. The defining trait of these wedges is their spring-loaded nature. The presence of an internal spring allows these wedges to absorb the backwards and downwards force exerted by chair users, allowing them to increase the force needed for the chair to reach its tipping point compared to a standard, non-spring-loaded wooden wedge. Testing data can be found further in the report detailing differences in tipping point forces using the adjustable doorstop compared to ones with alternative products and mockups.

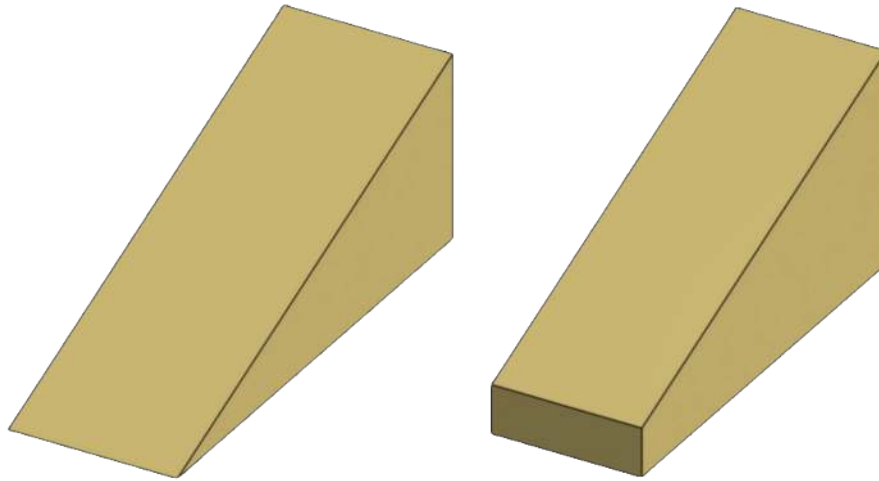


Figure 4: Short (left) and tall (right) wooden attachable bases

Although the spring-loaded door stops were powerful components to cushion force enacted on the chair, the angled nature of the backside of the dorm chairs made them not flush and parallel to the floor. Therefore, we took inspiration from our previous mockups, which used angled wooden blocks to attach to the chair legs, to create smaller wooden blocks that connect to the bottom of the doorstops to make the bottom of the entire system flush and parallel to the floor. This also allowed for size customization by constructing another set of wooden blocks with two extra inches of height, which are level with the 2 inch bed risers that can be attached to the front of the chair. Wood was an ideal material for this component, being lighter weight than a block of steel yet still being stable and durable. It was also a very straightforward process to shape a block of wood to fit the specified dimensions in our final prototype blueprint.

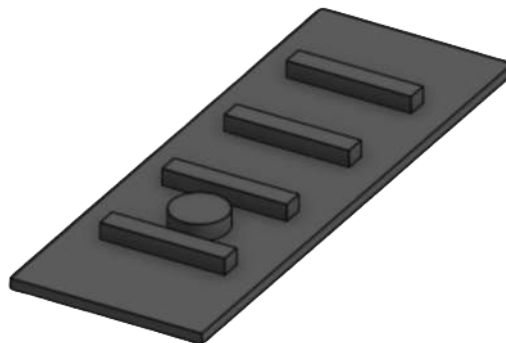


Figure 5: Rubber attachments found on surface of wooden blocks

Because of the decision to use an adjustable door stop attached to a smaller wooden block to increase stability (as opposed to a singular large block of wood), rubber attachments previously designed to fit onto the bottom of the door stops were glued onto the surfaces of the wooden

blocks through epoxy resin. The tops of these rubber attachments (Figure 5) have patterns that match with indents found on the bottoms of the door stops, allowing them to fit snugly onto the door stops while also allowing for smooth removal if a differently sized wooden block is desired.

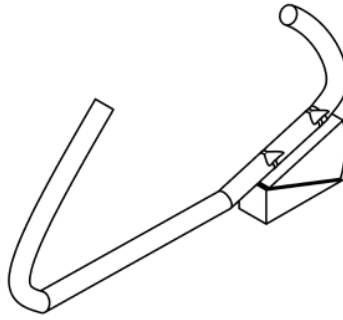


Figure 7: Our design with small size block attached

Seen in Figure 7, the smaller size wooden block takes up the unused space between the wedge riser attachment and the floor, making the back leg flush with the rest of the chair and adding stability without needing to raise the chair in any manner.

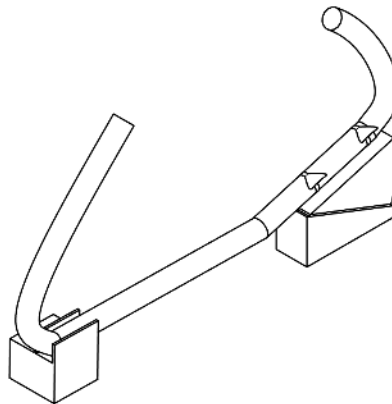


Figure 8: Our design with large size block and front leg risers attached

The larger wooden block provides a taller alternative for users who require more height on the chair. These ones add a two inch elevation to the chair when paired with our two adjustable screw clamp legs, which can be added to the front of the chair. A visual of this system can be seen in Figure 3 above.

This design circles around two vital subproblems discovered in our user interviewing: stability and adjustability.

Stability

To ensure maximized stability in our design, we utilize a large wedge with a spring inside, which helps to dampen the impact of a user pushing back on the chair when sitting down. This was proven to increase the quantity of force needed to reach the chair's tipping point, as the spring system cushions the impact of the force and slightly pushes the chair downwards. We also guarantee stability in the design through the metal hose clamps being used, which are significantly more durable than plastic alternatives while still being relatively easy to apply for users.

Adjustability

To allow users the option to adjust chair height, we implemented the aforementioned two size design. Through our user questionnaire, we found that some students felt the chair was too short, while others did not have any complaints on the height and were more concerned about other factors (such as stability). Therefore, our two size system would provide an effective compromise, allowing extra height for anyone who requires it and making it optional for those who do not.



Figure 9: Various view of the elevated block configuration

Force and feedback testing were conducted on the chair, providing valuable information on the efficacy of the WedgeRise design (Appendix G: Mockup Feedback and Testing Summaries). The table below holds data from force testing, in which the WedgeRise, other existing solutions, and a chair with no attachments were all tested for the Newtons of force required for the chair to reach its tipping point at various weight levels (added via sandbags). This testing demonstrates the stability in our design, which outperformed all existing products, as well as the control, in every weight level.

		Weight on Chair				
		0lb	42lb	90lb	122lb	150lb
Chair Add-ons	WedgeRise	40	70	170	255	265
	Wedges	12	25	45	100	150
	Control	25	55	70	125	178
	Bed Riser	25	50	75	175	225

Table 1: Force (N) Required to Reach Tipping Point of Chair at Different Weights

Feedback testing generated similar results, with users finding the chair very stable, comfortable, and easy to adjust height (Appendix G: Mockup Feedback and Testing Summaries).

Limitations and Next Steps

The WedgeRise addresses the two most significant issues with the Northwestern dorm chairs. The device reduces the chance that the user slips when sitting in the chair and allows the user to adjust the height in two distinct stages based on their preferred height. However, further refinements can improve both the stability and adjustability of the WedgeRise.

1. Adjustability

First, the device currently does not have a fully adjustable mechanism that allows the user to adjust to whatever height they desire, and must rely on the distinct stages designed by the manufacturer. For future iterations, other height adjustability mechanisms can be explored, such as the adjustability mechanism found in the Chair Riser Assembly (Appendix X: Patent Research Summary) or telescoping mechanisms found in office chairs.

2. Universalizability

In addition, the WedgeRise has only been designed to be installed on select Northwestern Dorm Chairs that have pipe style legs. Alternative attachment methods could be explored on the WedgeRise so it can be installed onto different styles of chairs, such as chair legs with a rectangular profile rather than a circular profile. For example, further testing could be done on

the clamp style mechanism found on the Uxcell Bed Riser to see if this clamp design could be retrofitted onto the WedgeRise.

3. Inclusivity

Finally, while a large number of Northwestern students participated in our user testing, our testing population was mainly limited to students from this university. Further testing could be done with different demographics such as younger children or older adults to see whether the WedgeRise can make dorm chairs more comfortable for those people as well. Nevertheless, Northwestern students are the primary user of these chairs, and their feedback holds the highest value in regard to our product since they are the primary user of this product.

Conclusion

The WedgeRise successfully improves the stability of the Northwestern dorm chairs and allows the height of these chairs to be adjustable. Each of the criteria for success outlined in the Project Definition (Appendix E: Project Definition) is addressed by the WedgeRise. Our device can be installed on Northwestern dorm chairs without specialized tools (Appendix X: Instructions for Use), has a relatively small footprint (The size is specified in the Design Concept and Rationale Section), is height adjustable through our swappable base mechanism and can hold up to X lbs (Insert weight testing data). In addition, we were able to retain some of the rocking motion that is currently present on the Northwestern dorm chairs through the use of the Adjustable Door Stop attached to the back legs.

Most importantly, the WedgeRise improves the stability of the Northwestern Dorm Chairs. The WedgeRise increases the force required to tilt the chair back by X lbs (Insert Force Testing Data) compared to the chair without any stabilizing mechanism. Through our user testing, on a scale from 1 to 5, the stability score from X Northwestern students was X points higher (Insert User Testing Data) when using the WedgeRise.

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<https://www.bifma.org/page/StandardsShortDesc>

Appendix

Appendix A: ALTERNATIVES MATRIX

Categorizing all Ideas:

<i>Can the user adjust the chair to a specific height?</i>	<i>Does it provide stability for the chair?</i>	<i>Does it make it more comfortable for the user to put their hands on the desk?</i>	<i>Will the device be securely attached? (Safe to use)</i>	<i>Would the device be a space-efficient and aesthetic solution?</i>
Telescopic Clamps Four adjustable risers Leg sleeve Additional risers on bottom of chair	Four adjustable risers Leg sleeve Chair leg caps	Rod with bands Chair Leg Sleeve Chair Platform Telescopic Clamps	Chair Leg Sleeve Chair Platform Chair leg caps	Rod with bands Chair Leg Sleeve Telescopic Clamps Additional risers on bottom of chair

Narrowing down to 4 Ideas:

Criteria (1-5) + Multiplier	Idea 1: Four adjustable risers	Idea 2: Leg Sleeve	Idea 3: Chair platform	Idea 4: Telescoping clamps
Adjustability (x5)	5	4	3	5
Comfortable (x4)	4	5	5	3
Easy to install/use (x4)	4	4	3	2
Durable (x2)	3	4	5	2
Stability (x5)	3	4	5	2
Aesthetic (x1)	3	4	2	4
Cost (x3)	3	3	4	3
Totals:	94	97	96	72

Appendix B: Need Finding Research Summary

Introduction

Our team was tasked with identifying a white space—an unmet user need that is frequently experienced—and developing a solution to address it. This report outlines the methodologies we used to gather over 110 unmet needs, as well as our systematic process for narrowing this list

down to one key need. We emphasized gathering diverse perspectives and minimizing bias through structured evaluation criteria.

Problem Generation

To cast as wide a net as possible, we employed four primary methods: surveys, interviews, public observation, and group brainstorming. The combination of all methods yielded over 110 unique needs.

1. Survey

We designed a short and approachable survey that included a brief explanation of our objective and a few open-ended questions about daily unmet needs. To maximize engagement, we kept the tone casual and all questions optional, so the responder could include whichever ideas resonated most with them. Additionally, we organized the survey into categories—Home, School, Transportation, Technology, and Miscellaneous—with subtext prompts (e.g., “Public transit? Biking? Weather? Tried solutions?”) to guide user reflection.

Survey Questions:

1. “Category 1: At Home (or In the Dorm) — Chores? Dorm annoyances? What solutions have you tried?”.
2. Category 2: School — Productivity? Studying/workspace? What solutions have you tried?
3. Category 3: Transportation — Public transit? Biking? Weather? What solutions have you tried?
4. Category 4: Technology — Phones? Ease of use? What solutions have you tried?
5. Category 5: Miscellaneous — Anything else? What solutions have you tried?

We distributed the survey to friends, family, and online groups, ultimately receiving 34 responses. We found the most success directly sending the survey to close family and friends, instead of sending to massive groupMe chats. While this broadened our reach, our sample remained limited to our extended networks.

2. Interviews

Several team members conducted informal interviews with friends and family using the survey questions as a guide. Team members approached their friends during casual conversation and posed broad questions about any needs or annoyances in their day-to-day life. This allowed us to ask follow-up questions and obtain richer detail regarding unmet needs and attempted solutions.

3. Public Observation

By observing public spaces on campus (e.g., Norris Center, Main Library), we identified pain points in students’ everyday experiences that may not have been self-reported in surveys or interviews.

4. Brainstorm

We supplemented our dataset through multiple group brainstorming sessions, focusing on personal frustrations, common student challenges, and non-student needs. We made a conscious effort to document real-time inconveniences throughout our day-to-day lives.

Problem Refinement

Once we had compiled our list of 110 needs, we engaged in a three-stage refinement process in order to get down to 25 Needs: Categorization, criteria rank, and market research.

1. Categorization

We used AI tools (ChatGPT, Gemini, and DeepSeek) to group the raw responses into thematic categories such as “Daily Living & Chores”, and subcategories like “Laundry & Clothes”.

ChatGPT [1] yielded the most coherent categorization when prompted, “Can you separate this list into different categories? Can you keep the ideas verbatim to how they were stated (and just put them under the categories) [Inserted list of 110 problems stated]”, to retain original phrasing while organizing by topic. This allowed us to quickly spot redundancy and overlapping issues.

2. Criteria Rank

Next, we developed a ranking system to evaluate the quality of each problem. After researching industry best practices and discussing as a team, we identified the five criteria in Table 1 below and assigned each one a weight based on its importance.

Table 1: Team ranking of filtering criteria for 100 problems

Rank	Criteria (scale of 1-10 points)	Multiplier
1	Feasibility in ten weeks	5
2	Problem relevance (Impact)	4
3	Team interest	3
4	Technical Complexity	2
5	Cost/resources	1

Feasibility in ten weeks was most important because of the scope of our class. Problem relevance was next because we wanted a product that would uphold the NSPE Code of Ethics for Engineers [2] and truly make an impact on the greatest number of users. Next, team interest was important because we wanted to work on a project that would keep us motivated and invested. Technical complexity followed—ideally, we wanted a solution that was impactful but required only modest development effort. Lastly, cost/resources were considered least important, since if a solution was feasible within ten weeks, resource needs could likely be addressed through DTC

department support or creative workarounds. Based on these priorities, we assigned multipliers to each criterion (from $\times 1$ to $\times 5$) to quantify their relative importance.

Each team member rated a subset of needs on a 1–10 scale for each criterion, as shown in Table 2. These scores were then multiplied by their respective weights, summed up, and sorted using Google Sheets. This produced a ranked list of needs based on total score. Although we aimed for objectivity, we noted that a 1–5 scale may have helped reduce subjective variation between scorers.

Table 2: Criteria Ranking Google Sheet

Problem	Feasibility (x5)	Problem relevance (x4)	Team interest (x3)	Technical Complexity (x2)	Cost/resources (x1)	Total Score
backpack getting soaked even w umbrella, pants getting soaked even with umbrella	9	10	8	8	6	131
Not enough places to put things despite wall space	10	8	7	8	9	128
It's hard to always carry an umbrella when it rains because afterwards, you have to put it away but it's wet so there's nowhere to store it	8	10	10	4	3	121
I like to wash dishes by hand. It takes a long time to wash dishes after each meal every day.	8	8	8	7	8	118
Dorm door slams shut loudly	9	7	8	6	9	118

The team reconvened to review the top 25 results. We checked for duplicates and condensed overly specific responses into more general statements. For example, “It’s hard to always carry an umbrella when it rains because afterwards, you have to put it away but it’s wet so there’s nowhere to store it” was refined to “Storing wet umbrellas”. This ensured clarity and consistency across our final selection. Through the process of categorization, criteria ranking, and double checking, we were able to condense our needs list to 25 most important ideas.

3. Market Research

After identifying the top 25 problems, we conducted more in-depth patent and market research (primarily through Amazon and Derwent Innovation) to explore existing solutions and identify gaps in knowledge or areas for improved user experience.

We divided the top 25 problems among the four team members, with each person responsible for researching the market landscape for a subset of problems. For example, for the issue of umbrellas not covering the entire body, we discovered commercially viable options such as silicon shoe covers and extended umbrella designs available on Amazon. In parallel, we found patents for umbrella supports in databases like Derwent Innovation.

Following this individual research phase, the team regrouped for a deliberation session. We presented our findings and discussed each idea in terms of similar criteria as the initial: existing market saturation, design feasibility, user impact, team interest, and relevance. Through this collaborative process, we narrowed the focus to one problem: Improving dorm rocking chairs as seen in Figure 3.



Figure 3: Northwestern Dorm Rocking Chairs

This particular problem stood out for several reasons. It appeared three times in our survey responses, and ranked #2 overall in our problem scoring matrix. It demonstrated high feasibility within the constraints of our course timeline. It serves a large and relevant user base as shown by the ample online forum response in Figure 4. Northwestern students in dormitories, as well as most college dorms in the US express frustration about these anti-suicide chairs making them “suicidal”. Finally, our research showed a lack of adequate existing solutions.

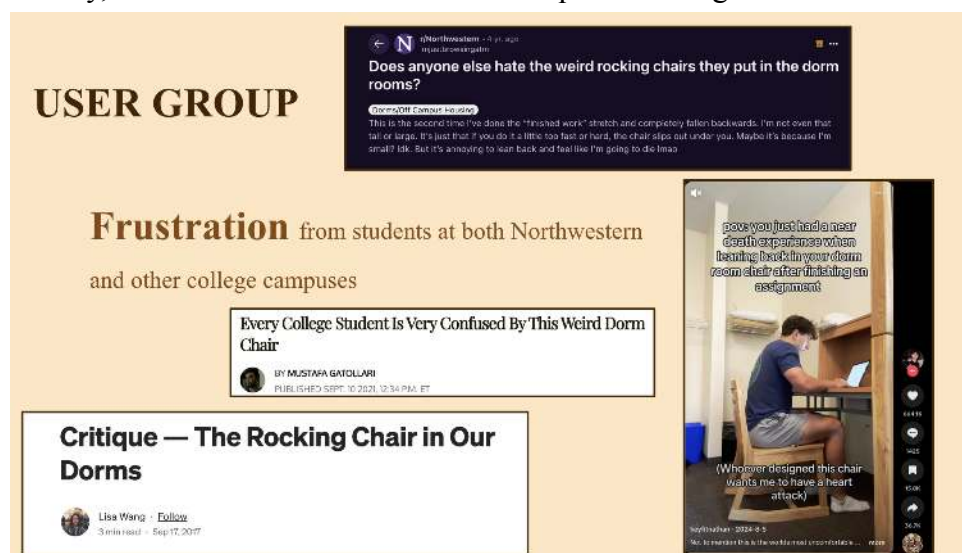


Figure 4: Online User Complaints of Dorm Chairs

As seen below in Figure 5 and 6, most commercially available solutions were designed for four-legged chairs or involved booster pads to improve comfort. Patent searches yielded a few

complex mechanical solutions for adjustable chair elevation, but many of these were associated with expired or inactive patents.



Figure 5: Amazon SONGMICS 4-Pack Furniture Risers [3]

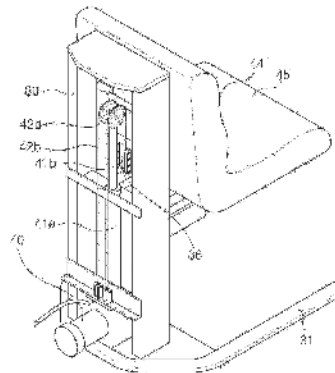


Figure 6: KR2439019B1 Patented Chair Adjuster [4]

Given its strong user relevance, technical simplicity, and limited competition in the current market, our team selected dorm rocking chair improvement as the final focus for the design project.

Conclusion

Our team was challenged to identify an unmet user need and design a meaningful solution. We employed a multi-method approach—surveying, interviewing, observing, and brainstorming—to generate over 110 user problems. We then applied structured criteria to rank and refine these needs, ultimately selecting one for further development: Unstable dorm chairs. With our final need identified, we are now prepared to move into the engineering design phase: ideating, prototyping, testing, and iterating our solution.

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Appendix C: Competitive Benchmarking Research Summary

Introduction

When our team decided to focus on improving the quality of dorm chairs at Northwestern, we searched across the internet for patents and products that aimed to solve this same problem. We took note of what we liked about each idea, and what we could improve upon with our design. This report summarizes our strategies and findings from our research.

Finding a Problem

Before we narrowed down our problem to dorm chairs, we took our top 25 problems and searched through a patent database and Amazon for solutions that were already made. This significantly helped us decide which direction to take, as a lot of our problems already had simple and accessible solutions. However, we didn’t find much in terms of solutions for the specific dorm chairs that we are tackling, so we chose to make one ourselves.

Problems with the chairs

When we gathered problems from our peers, there were many responses about the dorm chairs. But some complaints were focused on different aspects. One problem that seemed to have more available solutions was the lack of comfort and back support of these chairs. Back and bottom cushions like figures 1 and 2 are all over Amazon:



Figure 1: Ergonomic lumbar pillow



Figure 2: Fortem office chair cushion

Review: It is very comfortable and supportive. The product itself is of good quality. I'm very happy with it. -ZFB

These types of cushions work for some people that are only experiencing back pain, but our goal is to also increase stability and adjustability of the chair itself.

Our team decided to focus on an aspect of these chairs that was less easily solved: their stability. We searched for risers and adjustable clamps to add to these dorm chairs, but there was not much available for these chairs, which have one bar on each side that goes along the entire width of the chair as seen in figure 3.

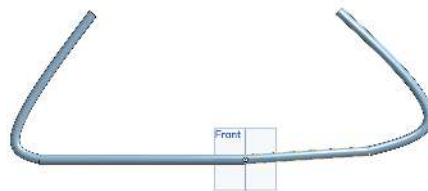


Figure 3: Approximate CAD design for shape of the bottom of dorm chair

We found a large amount of normal risers on Amazon, for couches and beds. But, because of the shape of these dorm chairs, those solutions would not work. As seen in figure 4, we also found a more complex riser that also had clamps:



Figure 4: Zeminior furniture risers with adjustable screw clamp

This solution seems like a step in the right direction with the clamps, but it still wouldn't work with the curvature of our dorm chair. Our team is thinking about including clamps in our solution, so we could look at this solution for inspiration. These risers also have felt bottoms so that nothing gets scratched, which we can also implement.

Conclusion

Based on our market research, there are a lot of solutions that could improve certain chairs and furniture. However, there is no widespread solution for making these dorm chairs more stable and adjustable. Moving forward we can take inspiration from adjustable risers with clamps and try to improve the design to fit our problem better.

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=1745951609&prefix=domr%2Bchair%2Bback%2Bcushion%2Caps%2C106&sr=8-5&th=1

[3] Zeminior, “Furniture Risers with Adjustable Screw Clamp,”

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dodzEeteuYHJllynpJioJdcchYLg539EpTMUaVYSX2YZ3Px53hjekjECK7aTps.mIgdxx0udlGo6
vMAlg5J0pqMTvMzGw8Bzkm3hiyNsUU&dib_tag=se&keywords=dorm%2Bchair%2Blegs&qid=1745948754&prefix=dorm%2Bchair%2Blegs%2Caps%2C114&sr=8-16&th=1](https://www.amazon.com/ZEMINIOR-Furniture-Adjustable-Supports-Extenders/dp/B0DJ341968/ref=sr_1_16?crid=2S08ODJURGIM6&dib=eyJ2IjoiMSJ9.h8r5yzt6i8FW1iHEuHl05OCHxt71N5JMRno-MgXxeS32tNtrvtwHKmo1FFCs3yWLEtBEaWJ4a3RLdo0RLZ7kZf7xmvWICRNm9EpTh5QEHxITX_NaMYsPiNOcCcTiu46Andc6X01fdm71sFJuRSbx6P4G_ocSrePcbsbtQbEWXJyDOEHRfHbW1D40vQLjyLOFsT0_BPxm3pP9E3Fn-pJt6bWfJf-Ry_wL6vo67XFgCAAYce
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vMAlg5J0pqMTvMzGw8Bzkm3hiyNsUU&dib_tag=se&keywords=dorm%2Bchair%2Blegs&qid=1745948754&prefix=dorm%2Bchair%2Blegs%2Caps%2C114&sr=8-16&th=1)

Appendix D: Patent Research Summary

Primary Author: Kevin Lin

Team Members: Aydn Calhoun, Dorothy Zhang, Pranav Neupan

Introduction

For our DTC Whitespace project, we decided on improving the dorm chairs at Northwestern University; specifically, our goal is to make the chair stable and height adjustable. This patent research summary includes the current solutions we used to help us generate ideas for our potential solution. Many of the examined patents provide both stability and adjustability for the types of chairs that they were designed for; although, they are unlikely to work with the Northwestern dorm chairs. The status of each patent is listed below each patent along with a description of the product and a short analysis on how we used the patent in our design research.

Elevating Chair

The Elevating Chair (Figure 1) is a mechanism to dynamically raise or lower the height of a chair using a system of pulleys and motors [1].

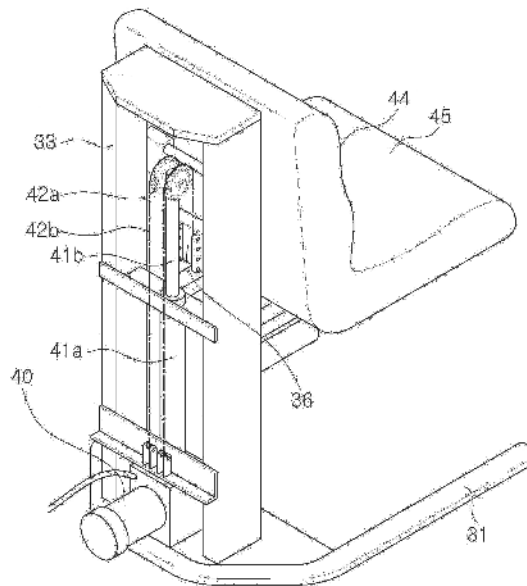


Figure 1: Drawing of the large adjustable mechanism for the chair (KR2439019B1) [1]

Status: Alive

Date Filed: 2022-08-31

Date of Expiration: 2042-04-15

Description:

This product is designed to help elderly people raise or lower their chair and assist them in getting up out of their chair. A pulley mechanism on the back of the chair allows the chair to adjust to different heights with a motor. The large metal base allows the chair assembly to be stable, even with a large moving load applied onto it.

Analysis:

There are some products that are designed to stabilize chairs and change their elevation. However, the large electrical mechanism on the back of the chair is expensive to produce. Additionally, the assembly is hard to move back and forth which is an essential component of the dorm chair since students are often shifting the chair towards and away from their desk.

Height Adjustment Structure for Chair or Table

The furniture height adjuster (Figure 2) is a product that slides under the base of a piece of small furniture and uses its telescoping legs to adjust the height of the furniture to the desired height [2].

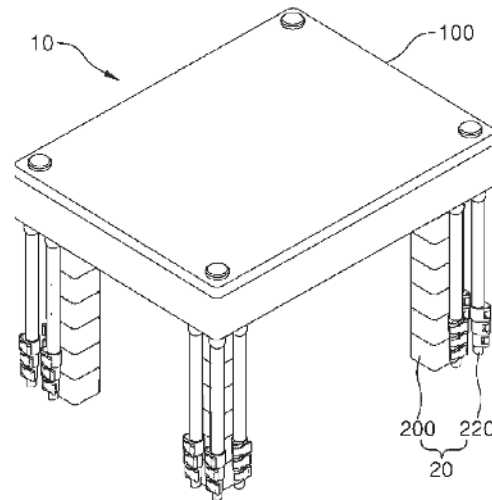


Figure 2: Diagram of the Height Adjustment Structure (KR2024058300A) [2]

Status: Alive

Date Filed: 2024-05-03

Date of Expiration: N/A

Description:

The design features a large flat base attached to a system of clamps on each leg designed to lock the retractable leg into a specific position of the desired height. Each leg also contains a stack of blocks, designed to increase the contact area of each leg with the group for a stable base.

Analysis:

This product allows for a large range of adjustability when placed under a table or small chair. However, the relatively narrow and flat base does not accommodate the structure of Northwestern Dorm Chairs and the design of the legs does not seem to accommodate the large forces that may occur when using this product as a base for a chair.

Chair Riser Assembly

This chair riser assembly (Figure 3) attaches to the base of large chairs and uses a system of sliding feet that can be locked into place to adjust the height of the chair [3].

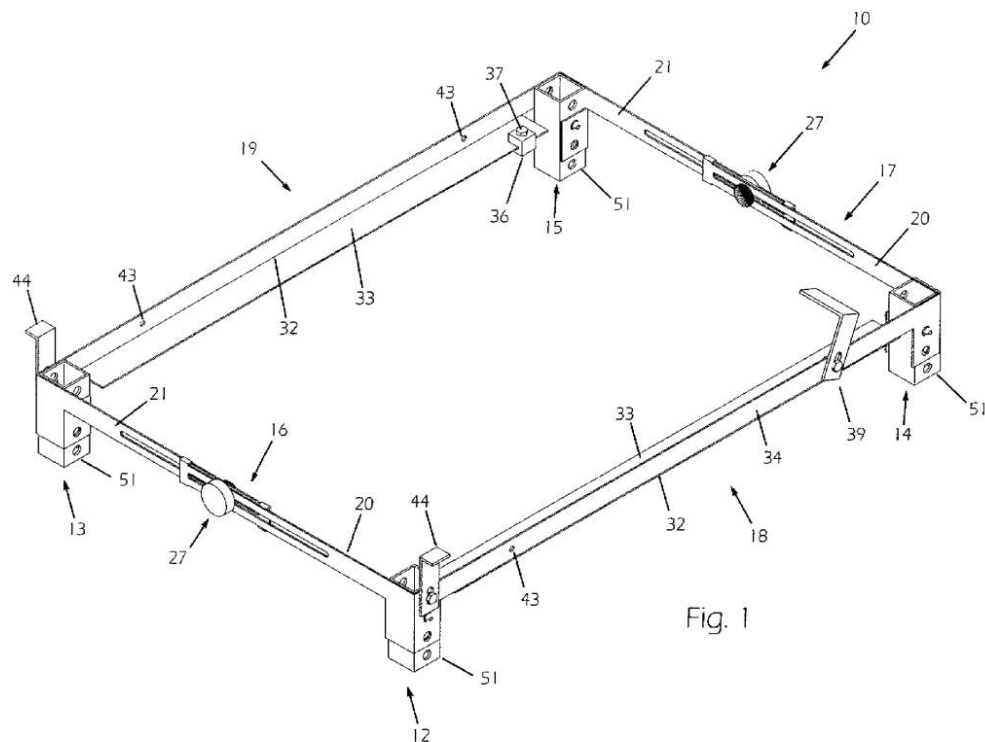


Figure 3: Large mechanism to raise chairs (US9364093B2) [3]

Status: Alive

Date Filed: 2015-05-08

Date of Expiration: 2035-05-08

Description:

This chair riser serves a similar function to the furniture height adjuster with a larger and shorter base. However, this design includes dials on the front and the back that allow it to fit on the base of a greater range of chairs. Each leg has two components: a base and a sliding foot. The sliding foot has a set of holes that allows it to be bolted to the frame of the device and adjust the height. Finally, this design also includes rails that extend above the base of the device to create a secure attachment to the chair.

Analysis:

While this chair riser does allow for easy adjustability using its mechanism of the sliding foot, the design still faces the same issue as previous designs when considering the shape of the Northwestern dorm chairs. The metal protrusions on the side of the device that are meant to hold

onto large reclining chairs will likely not maintain a secure grip on the curved tube structure on the legs of the dorm chairs.

Chair Riser

This static chair riser (Figure 4) is a simple assembly designed to attach to the bottom of chairs and help raise the base of the chair [4].

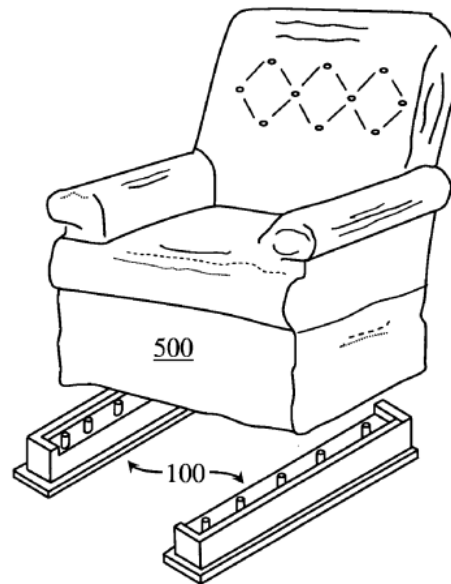


Figure 4: The chair stabilizer being attached to the bottom of a large chair (US6929328B1) [4]

Status: Dead

Date Filed: 2004-01-21

Date of Expiration: 2024-01-21

Description:

The purpose of this product is to help elderly citizens get out of their chairs since the low height of some chairs make it difficult for them to get up. The pegs that line each side of the device are designed to attach to create a stable attachment to the bottom of the chair while also being easily removable for transportation.

Analysis:

This product adequately supports the base of large chairs such as the reclining chair depicted in Figure 4 due to the 4 pegs and large surface area that attaches to the chair. While this does provide a stable platform for the chair, the lack of incline and adjustability make it difficult to solve the issues presented for the dorm chairs.

Conclusion

In general, many of the patents that we examined during the patent research process were not suited for use on the Northwestern dorm chairs, mainly due to the curved, tube structure of the dorm chairs. However, these patents contained design elements that may assist us in implementing our own design, such as the mechanism for adjustability present on the Height Adjustment Structure [2] and the Chair Riser Assembly [3]. Moving forward, we will keep the benefits of each of these structures in mind when designing a product that can be used to elevate and stabilize the Northwestern dorm chairs.

References

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[3] Chair Riser Assembly, by Ronald L. Williams, Anthony J. Nowakowski. (Jun. 14, 2015). US9364093B2 [Online] Available:

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Appendix E: DESIGN REVIEW SUMMARY

Introduction

During our design review, we presented our first mock-up, four wooden blocks with plastic clamps that connect to the bottom of the chair. We showed our audience the methods we used to benchmark and test our mock-up, and showed other products found on amazon. Finally, we discussed the pros and cons of our current design, as well as next steps to improve it. Our feedback was generally positive, with users testing it out and feeling stable. However, we did get many suggestions to look at further.

Current Design Feedback

- Users felt generally **stable** in the chair
- Wood blocks felt a little uneven, with a user feeling that one of the **front blocks were not fully contacting the ground**
- **Clamps** were made of **weak material** and did not hold up as well as needed

Future Design Feedback

- Multiple **small wood blocks** that could be **stacked** to adjust height
- **Remove front blocks** entirely and explore a **hinge / piston device** that goes on the back
- Add **extra weight on the front** side of the chair to further prevent tipping
- Student asked about having the **attachments further on the fronts/backs of chairs** to advance stability
- **Softer attachment bottoms** to cushion contact with ground
 - Felt
- Proposal to add **elevators to only back legs**
- No feedback on other materials to try to use → follow through with wood?

Appendix F: Project Definition

Project Name: Dorm Chair Stabilizer & Enhancer

Team Members: Aydn Calhoun, Dorothy Zhang, Kevin Lin, Pranav Neupane

Date: 5/19/2025

Version: 3

Mission Statement

Design a retrofittable tubular attachment for college students with rocking dorm chairs, who want a stable and height adjustable dorm chair. This product should address issues of rocking while leaning back, base scraping the floor, and safety standing on the chair.

Project Deliverables

- **Final Report, *digital copy***
 - **Infomercial and Final Presentation**
 - **Final Prototype**
-

Design Constraints

- Maximum budget is \$150
 - Time constraint - must be able to be completed before the quarter ends
 - Follows any relevant federal regulations
-

Users and Stakeholders

Primary Users:

- Northwestern students who live in dorms with the rocky, tubular base chairs.
- Other college students who sit in chairs with a similar stature/

Secondary Stakeholders and Users:

- Friends of family of the students who will sit in the chairs.
-

Illustrative User Scenario

This scenario is based on the stakeholder interview, in-person observation, and personal experience from team members.

After a long day filled with classes and work, Bobby, a first-year student at Northwestern University, finally returns to his dorm room hoping to relax and complete tedious class assignments. As he settles into his chair, he notices scratch marks on the floor caused by repeated movement of the chair. Glancing up, he realizes that a poster has fallen from the wall. Attempting to re-stick it, he then stands on the unstable chair in an effort to reach higher, but loses his balance, falls, and injures himself against the desk.

Frustrated and in pain, Bobby sits back in his chair to recover. However, the chair, lacking proper support and stability, tips backward, causing him to fall again—this time striking the wall behind

him. Now physically hurt, emotionally drained, and overwhelmed by the cumulative stress of the day, Bobby is going nuts.

Table 1: Requirements for the scope of the project

Needs	Metrics	Units	Ideal Value	Allow-able Value	Justification
Durable and low maintenance	Lifespan before structural failure or performance degradation	Years	15	5	College furniture is typically used over multiple student cohorts. Industry furniture (e.g., Herman Miller) sets lifespan targets of 10–15 years. Given expected rough use in dorms, 5 years is acceptable for prototype
Retrofit Stability	Ease of installation by user without specialized tools	1-5	1	3	Based on user testing feedback, most students do not have access to power tools. A tool-free installation (rating 1) is preferred. Installation steps > 3 were found to reduce user compliance.
Size	Additional floor footprint introduced	ft ²	0	1	Dorm rooms are typically under 150 ft ² . Maximizing compactness minimizes obstruction. Ideal footprint aims for no added protrusion beyond existing chair.
Height adjustability	Range of vertical adjustment possible for chair seat	in	>5	>3	ADA recommendations for accessible seating suggest height flexibility to accommodate users of varying sizes (CDC 2021) ¹ . Survey results indicated that students prefer ergonomic adjustment to desk height.

¹ <https://www.access-board.gov/ada/>

Stability when use	Tipping resistance and sway under normal and moderate load	1-5	5	4	Interviews and injury reports indicated a significant user pain point around tipping (Appendix A). ASTM F2613-21 standard for seating requires minimal deflection during use. ²
Maximum weight capacity	Structural load bearing without deformation or failure	lbs	300+	275+	ANSI/BIFMA X5.1 standard for general-purpose chairs sets the minimum weight limit at 275 lbs. To support a diverse user population, a minimum is set. ³

² <https://www.intertek.com/products-retail/insight-bulletins/2021/cpsc-adopts-astm-f2613-21/>

³ <https://www.bifma.org/page/StandardsShortDesc>

APPENDIX G: MOCKUP FEEDBACK AND TESTING SUMMARIES

Introduction

On May 8th, Aydn, Dorothy, Kevin, and Pranav (the full group) met at the Shop to conduct mockup testing with our initial mockup, consisting of four wooden blocks shaped to fit into the angled portions of the two chair legs to ensure stability and eliminate rocking. Our team intended to determine the feasibility of this design, both aesthetically and practically, as a viable solution to the various problems related to Northwestern dorm chairs that we hope to solve. This session consisted of a general test of the mockup's fittingness on the chair and the chair's rocking ability with it. Observations were also made regarding the comfort of the chair with the mockup applied.

On June 5th, mockup testing with a refined design was conducted at the Shop by Aydn, Dorothy, Kevin, and Pranav (the full group) with hopes of validating the mockup's stability and ability to increase force needed to reach a chair's tipping point. This was coupled with a feedback session with 14 volunteers inside Ford, rating the chair on various subjective scales.

Methodology

May 8th Mockup Testing



Figure 1: Angled backside attachment of first mockup

1. **Installed our mockup** (Figure 1), consisting of four wooden attachments, onto the legs of the dorm chair brought to the Shop by DTC Section 9



Figure 2: Team member (Kevin) sitting on chair with mockup

2. Team members took turns sitting in the chair and **applying our “standard forces”**, which we defined as the extent to which we expect to push our chair forward or backward when sitting (Figure 2)
3. Team members each **pushed chair to its maximum tipping point** by having another person on the back side holding the chair to prevent user from falling
4. Repeated Steps 2 and 3 **without the attachment installed**
5. **Discuss our individual observations** and feelings of the chair’s stability with our mockup versus without it

June 5th Force Testing

1. **Installed the short version of our updated mockup/prototype** onto the chair legs of the dorm chair brought to the Shop by DTC Section 9
 - a. The tall version, which uses additional parts, would have an equivalent tipping point, so we decide to only test one (as the data would be equal)
2. **Push the chair backwards with a force meter** until it reaches its tipping point
3. **Record the max force** indicated by force meter during this process



Figure 3: Force meter testing on chair with sandbag weights

4. **Repeat steps 2 and 3 with incremental addition of sandbags** until it reaches four sandbags, testing force needed to reach tipping point at each sandbag quantity (Figure 3)



Figure 4: Visual of chair with wedge attachment

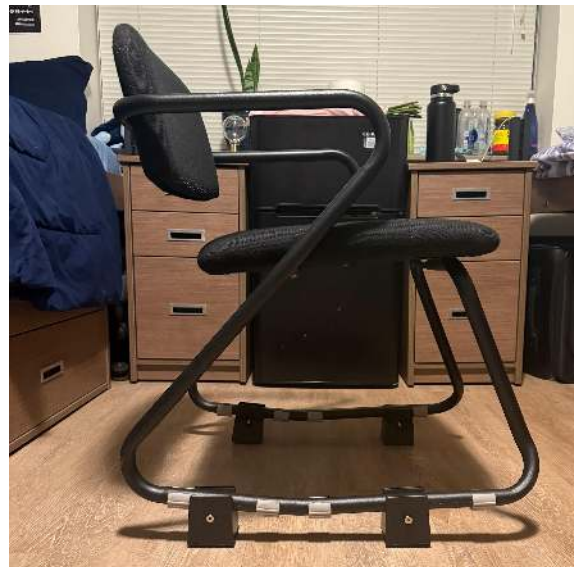


Figure 5: Visual of chair with bed riser attachment

5. **Repeat Steps 2-4** with no attachment on chair, with wedge attachment (Figure 4), and with bed riser (Figure 5)

June 5th Feedback Testing

1. **Brought to the entrance of Ford** our chair with the final mockup/prototype attached



Figure 6: Volunteer testing chair with our mockup

2. **Sought volunteers to test** the chair and **give ratings** from 1-5 on different criteria (Figure 6)

Results and Discussion

Our May 8th testing of the developed mockup was successful and brought valuable information for the team. We found that:

- The design could **support our weights and light forces both forwards and backward**
- **Tipping points were still reachable** with “standard force” by two of the four testers
- The chair **tipped easier than expected to the left and right**; this was a difficult value to measure quantitatively, but we were able to qualitatively conclude that the mockup design needed much more sideways stability.
- The plastic clamps connecting the design to the chair were **very fragile and began to break**

The results of this initial testing told us that we would need to find a way to **increase the force needed for reaching the tipping point** of the chair and more **securely fasten the design onto the chair** while not sacrificing the other aspects of the design, which we found to be effective.

Following this, we made adjustments to our design, most importantly adding a **spring-loaded door stop to the back** (dampening any force applied to the chair to hopefully increase force needed for tipping point) and **using metal hose clamps to securely fasten the design** onto the chair.

The June 5th force and feedback testing with this design brought the following data:

Table 1: Force (N) Required to Reach Tipping Point of Chair						
		Weight on Chair				
		0lb	42lb	90lb	122lb	150lb
Chair Add-ons	WedgeRise	40	70	170	255	265
	Wedges	12	25	45	100	150
	Control	25	55	70	125	178
	Bed Riser	25	50	75	175	225

Our force testing confirmed the efficacy of our new design in drastically increasing the force required to reach the tipping point on the chair at the weight ranges we were able to test. This was a dramatic improvement from the previous mockup, whose force testing data reflects that of the bed riser add-on.

We hope to conduct future force testing with a wider and higher range of weights in the future to ensure the design can sustain heavier loads and still maintain strong tipping point force values.

Table 2: User Testing Average Rankings on 1-5 Scale for Different Criteria					
Stability	Comfort	Aesthetic	Durability	Adjustability	Total Rating
4.695	4.62	3.65	3.54	4.69	4.24

Our feedback testing brought similar results, demonstrating an **evident strength in stability, height adjustability, and overall comfort.**

However, rankings were proportionally **lower in durability and aesthetics.** Multiple users recommended **applying black spray paint to the wooden part** of our mockup to make the color

more flush with the rest of the mockup (and with the chair), and others recommended to **bolt the clamps onto the mockup** to maximize durability.

Appendix H: Ethical Considerations Summary

Primary Author: Kevin Lin

Team Members: Aydn Calhoun, Dorothy Zhang, Pranav Neupane

Introduction

Our team created a whitespace project focused on improving the dorm chairs at Northwestern University. Specifically, we sought to make these dorm chairs more stable and height adjustable to accommodate the preferences of a wider range of students. This appendix will contain the ethical considerations taken during the design process of our device, using the NSPE Code of Ethics [1] as reference. This appendix will be split into 5 sections: development, manufacturing, user impact, social impact, and end of life. These sections were chosen based on the product management timeline [2], modified to include the sections that are relevant to this project.

Development

In every product designed to be marketed and sold to the general public, engineers must follow one of the most basic principles outlined in the NSPE Code of Ethics, which is to “hold paramount the safety, health, and welfare of the public” [1]. Since our project is designed to be marketed towards Northwestern University students and potentially students from other universities, we considered possible safety and ethical concerns that are associated with designing a product for the public.

The greatest concern of our project is the stability of the chair, since a failure in this regard may lead to the user falling out of the chair, leading to severe injuries to the head and other parts of the body. Therefore, we focused on the stability of the product when designing the product and the performance tests. In addition, the product must be safe for the consumer to use, meaning that it must remain safe even during unexpected contact such as when the user accidentally bumps into the device with their foot or other body parts.

As part of the development of the product, we tested the device on users to measure their satisfaction in our product. However, this may lead to unexpected injury; thus, we must take the necessary safety precautions and consent from the user before conducting any user testing. By increasing the stability of the chair, we hope to make the dorm chairs at Northwestern University less prone to fall and protect the health of our students.

Fabrication and Manufacturing

During the fabrication of our mockup designs, the materials used in our product must be strong enough to endure nearly any possible stress that could be reasonably placed on it. This means that the product must be able to hold the weight of any Northwestern student, along with the weight of the chair and any accessories that the user may want to put on the chair. Furthermore, the material must remain durable over time so the product does not unexpectedly fail and cause harm to the user. While the design should not incorporate materials that are unreasonably expensive, the cost of using less durable materials can potentially compromise the

health of the user. Therefore, strong and durable materials such as wood and steel were mostly used in the design and testing of the product as opposed to weaker materials such as 3D printed plastics.

In future manufacturing of our design, we must also consider the humanitarian and environmental impacts of our product. When considering the features of our final design, we must consider how current manufacturing processes can produce these types of products, and whether some features are necessary at all to reduce the labor and environmental impact during the manufacturing of our product.

User Impact

The impact of the product on the user relates to both the health and comfort of the user. Our device seeks to protect the health of any student who uses the dorm chairs at Northwestern University by reducing the risk of falling when leaning on the chair. Our product must have positive user impact in this regard, since failure may lead to severe health complications if a user falls out of the chair. Furthermore, our product intends to improve the comfort of the chairs by raising the chairs to a height where the user can comfortably work. A positive user experience in the comfort of the chair can allow students to be more productive while working at their desk and potentially avoid long term health complications related to posture.

Social Impact

The primary social impact of our device is improving access and inclusivity when working at dorms. Currently, there are many students of different heights and body shapes that attend Northwestern University and due to the lack of adjustability of the dorm chairs, some students find it uncomfortable to work at their dorms. Allowing students to adjust their chairs to their desired height gives more students access to better working environments. As mentioned in previous sections, the device also promotes safety by reducing the risk of falling out of their chair.

End of Life

During our design process, we considered how the device can be installed and uninstalled by the user, along with how the user can store the device when not in use. A small footprint was a key component in our design to allow students to easily store the device in the room and reduce the amount of material used in construction of the device. Furthermore, sustainable materials were preferred in the construction of our device, such as wood and metal as opposed to non-recyclable materials such as plastic.

Conclusion

We will hold the safety of the user as the most important ethical consideration during the design of our product especially as it relates to the stability of the chair, along with the privacy and comfort of the user as outlined in this summary.

References

- [1] National Society of Professional Engineers, *Code of Ethics for Engineers*, NSPE Publication #1102, July 2019. [Online]. Available:
<https://www.nspe.org/sites/default/files/resources/pdfs/Ethics/CodeofEthics/NSPECodeofEthicsforEngineers.pdfNSPE+3NSPE+3Course Sidekick+3>
- [2] Investopedia, "Product Lifecycle Management (PLM): Definition, Benefits, History," *Investopedia*, Nov. 29, 2010. [Online]. Available:
<https://www.investopedia.com/terms/p/product-life-cycle-management.asp>

Appendix I: WedgeRise: Instructions for Construction

Primary Author: Kevin Lin

The WedgeRise is a device that attaches to the rear of the Northwestern dorm chairs to provide greater stability and range of height adjustability. The Wedge Rise is constructed of 4 main components: The worm gear pipe clamp, the adjustable doorstop, the rubber attachment and the wood base as shown in Figure 1.

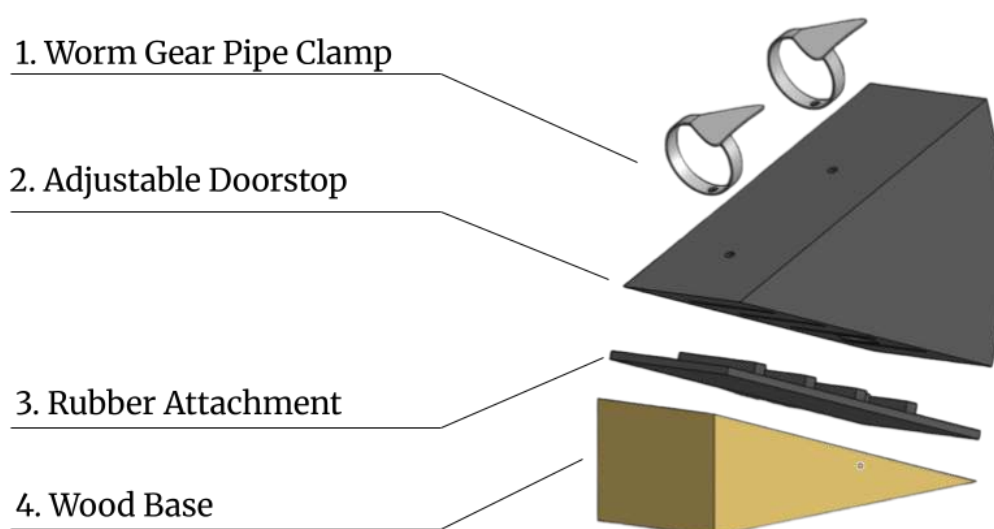


Figure 1: Exploded View of the components of WedgeRise

High Configuration

To create the WedgeRise with a taller base (2 inch rise), complete optional steps 1a and 2 under [Section 2: Fabricating the Wood Base](#). Additionally, purchase the Uxcell 2" bed risers under Table 1: Materials For Construction

Materials For Construction

The following table details the materials required from construction

Table 1: Materials For Construction

Material	Specification	Quantity	Link	Cost
2x4 Wood Plank	1 ½ '' x 3 ½ ''x 96''	1	Home Depot	\$4.98
Worm Gear Hose Clamp	1"-2" Diameter	4	Amazon	\$9.59
Adjustable Door Stopper	3'' Height	2	Amazon	\$19.98
Wood Glue	-	1	Home Depot	\$3.57
Epoxy Glue	-	1	Amazon	\$5.97
Wood Screw	#6 x 1/2"	2	Amazon	\$4.79
Machine Screw	M3 x 8mm	4	Amazon	\$7.99
(Optional) Uxcell Bed Riser	2''	2	Amazon	\$24.99

The following tools are required for construction

- Bandsaw
- Screwdriver
- Ruler or other measuring device
- Drill with 3mm drill bit

Preparing the Adjustable Door Stopper

1. Remove the rubber grips from the top and the bottom of the Adjustable Door Stopper



Figure 1: Adjustable Door Stopper with rubber grips removed

2. Use the Drill to create two 3mm holes approximately 2 cm and 8 cm from the base of the door stopper

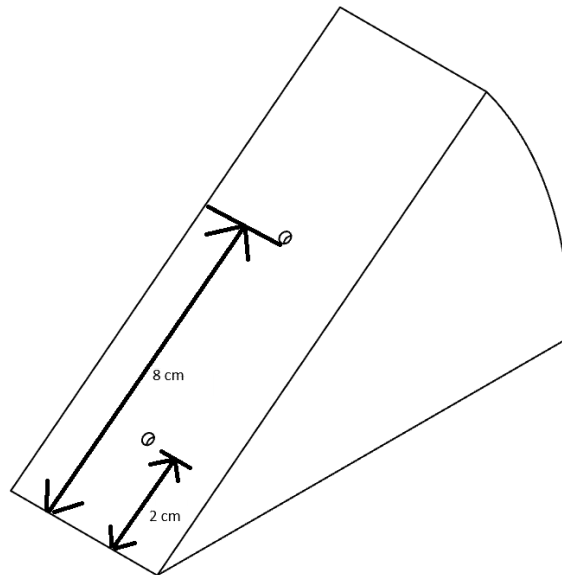


Figure 2: Isometric sketch of placement of holes on the Adjustable Door Stopper

3. Get 2 Worm Gear Hose Clamps and use a hand drill to drill a 3mm hole on the base of each clamp. Ensure that the hole is positioned on the solid metal portion of the clamp.

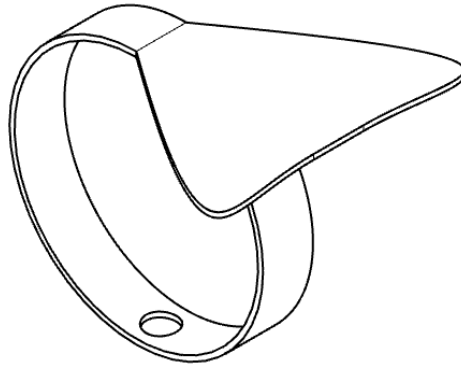


Figure 3: Isometric sketch of hole drilled into clamp

4. Place the two clamps above the holes on the door stopper and screw in the m3 machine screws, placing a nut behind the screw to secure it.

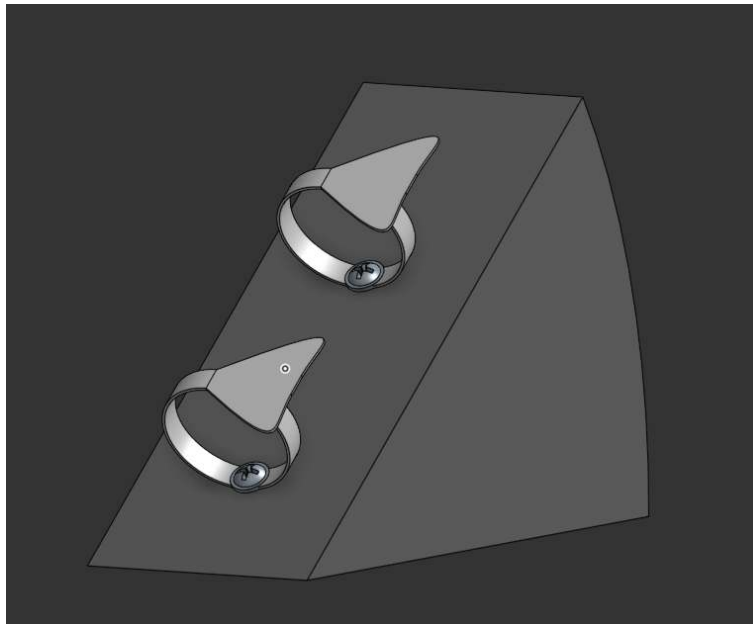
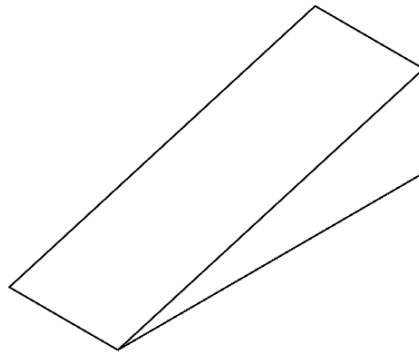


Figure 4: 3D Render of clamps attached to door stop using m3 machine screws

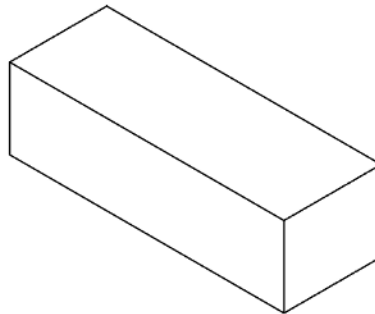
Fabricating the Wood Base

1.
 - a. For the smaller base, cut out a triangular block of wood that is 13cm x 4.6cm x 3.8cm, where the height of the wood is 0 at one end, and 3.8cm at the other end.



*Figure 5: 3D render of triangular block of wood.
Dimensions: 13cm x 4.6cm x 3.8cm*

- b. (Optional) For the taller base, use the bandsaw to cut out an additional rectangular 13cm x 4.6cm x 3.8cm block of wood.



*Figure 6: 3D render of rectangular block of wood
Dimensions: 13cm x 4.6cm x 3.8cm*

2. (Optional) Use wood glue to glue the triangular block of wood onto the rectangular block of wood.



Figure 6: Tall base constructed of triangular and rectangular blocks of wood, attached with wood glue

3. Put epoxy on the top of the wood base and screw in one of the rubber feet from the door stopper with a screw near the top of the wood in the orientation shown in the image



Figure 7: Rubber Feet attached to top of wood base using a single wood screw and epoxy

4. Clamp the shot end of the wood base with the rubber feet attached and let epoxy dry for 24 hours



Figure 8: Clamp attached to wood base used to hold rubber feet in place while epoxy dries

Final Assembly

Take the wood assembly and insert the rubber feet onto the underside of the adjustable door stopper.

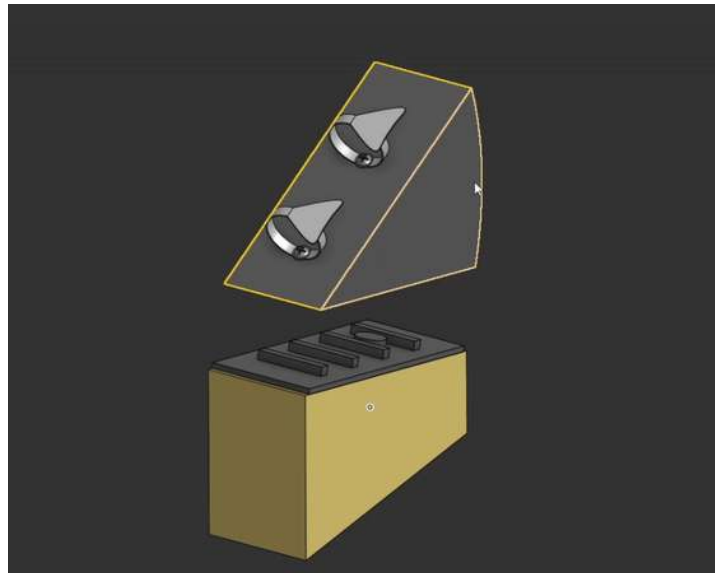


Figure 9: 3D simulation of attaching the adjustable door stopper to its rubber feet

The WedgeRise is now fully assembled and ready to be installed onto a dorm chair.

Appendix J: INSTRUCTIONS FOR USE

The WedgeRise is a retrofitted attachment for the backs of Northwestern University dorm chairs that prevents the chairs' characteristic rocking nature and provides the stability of a standard four-legged chair for users. Using a spring-loaded doorstop at the top of the product, the WedgeRise also helps significantly reduce the chair's tipping point by dampening the impact of downwards/backwards force upon the chair, drastically reducing users' chances of falling out. The WedgeRise also comes with two different sized attachments: one that makes the product's bottom flush with the front part of the chair legs (making it fully stable), and one that adds two inches of height and is complemented with two bed risers that are attached to the front legs (making the chair taller for shorter users).

Instructions for With the Small Block Attachments

1. **Position** door stop attachment

to the back legs of the chairs, putting the clamps between the chair legs. The two clamps should be on opposite sides of the white plastic wedge on the back of the chair. Have the chair upside down to ensure a smoother installation process.



Figure 1: Positioning door stop onto chair leg

2. Clamp on the spring-loaded attachment

wedge by screwing the wing nuts clockwise until met with significant resistance (Figure 2). The clamps must be tightly onto the chair with no empty space between the clamp's inner diameters and the chair leg.

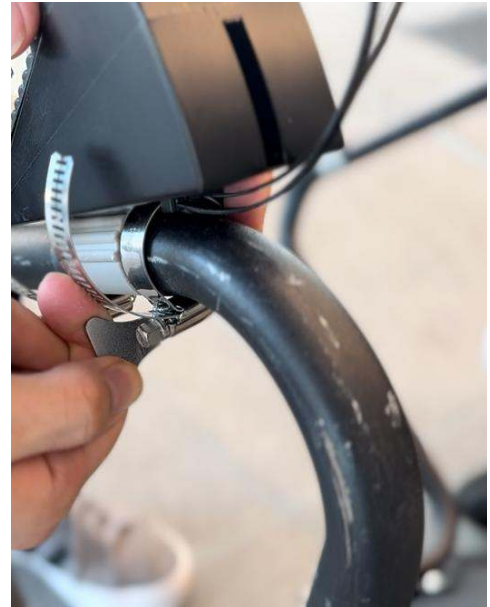


Figure 2: Tightening clamps on chair leg

3. Fasten a small wooden block onto the bottom of each clamped wedge. The rubber tops on the wooden blocks have attachments on their surfaces that will fit securely into equally shaped indents on the bottoms of the clamped wedges (Figure 3).



*Figure 3: Rubber attachment
and the associated indents on
wooden block*



Figure 4a, 4b: Attaching the small wooden block (4a) and product post-attachment (4b)

Ensure that the rubber surface is entirely flush with the clamped wedge following attachment. If not, then it is not surely fastened and should be pressed in more. The final product after this step should appear as Figure 4b.

4. **Reposition** chair to its normal configuration. It is now ready for use!

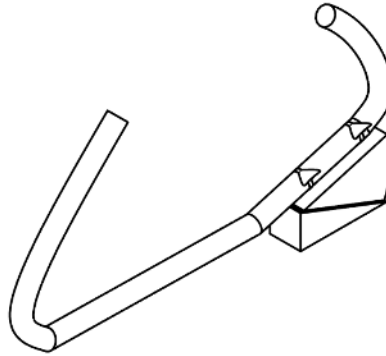


Figure 5: Final product after WedgeRise attachment (size small)

Instructions for with the Large Wooden Attachments

1. Position door stop attachment

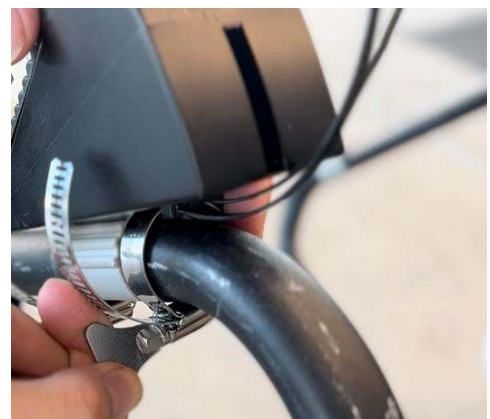
to the back legs of the chairs, putting the clamps between the chair legs. The two clamps should be on opposite sides of the white plastic wedge on the back of the chair. Have the chair upside down to ensure a smoother installation process.



Figure 1: Positioning door stop onto chair leg

2. Clamp on the spring-loaded attachment

wedge by screwing the wing nuts clockwise until met



with significant resistance (Figure 1b). The clamps must be tightly onto the chair with no empty space between the clamp's inner diameters and the chair leg.

Figure 2: Tightening clamps on chair leg

3. Fasten a small wooden block onto the bottom of each clamped wedge. The rubber tops on the wooden blocks have attachments on their surfaces that will fit securely into equally shaped indents on the bottoms of the clamped wedges (Figure __).



Figure 3: Rubber attachment and the associated indents on wooden block



Figure 5a, 5b: Attaching the large wooden block (4a) and product post-attachment (4b)

Ensure that the rubber surface is entirely flush with the clamped wedge following attachment. If not, then it is not surely fastened and should be pressed in more. The final product after this step should appear as Figure 5.

4. **Reposition chair** to its normal configuration. It is now ready for use!

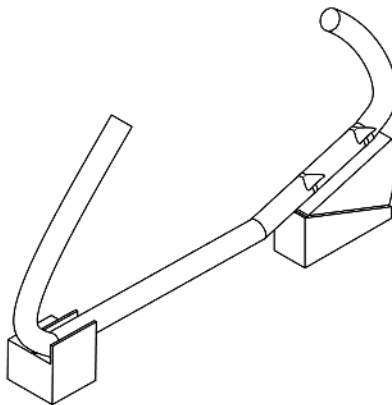


Figure 6: Final product after WedgeRise attachment (size small)

Appendix K: Bill of Materials

Date	Qty	Description	Specs	Supplier	Link	Total Price	Notes
4/29/25	1	Clamp to attach to chair leg	1"	HJ Garden	Amazon Link	\$8.99	Est arrival: 5/2
5/1/25	1	Adjustable clamp to elevate furniture	Furniture Thickness 0.39-1.96"	uxcell	Amazon Link	\$24.99	Est arrival: 5/2
5/1/25	1	Angled rubber wedges to put under chair on the angled portion of leg	3.15"-1.6"- .32"		Amazon Link	\$9.98	Est arrival: 5/3
5/6/25	1	Adjustable Gas Lift Cylinder for Office Chair Stool Replacement	Antlu 14" to 22" (8" Stroke)	antlu	Amazon Link	\$23	Est arrival: 5/8
5/20/25	1	1 1/2 inch Hose Clamp x10	Antlu 14" to 22" (8" Stroke)	sumprom	Amazon Link	\$10	Est arrival: 5/22
5/21/25	2	Heavy Duty Door Stopper with adjustable height via spring	3" Maximum Height	Yeegras	Amazon Link	\$20	Est arrival: 5/25
Total Purchase Amount						\$96.96	